



Yahui Li

Senior Undergraduate Student

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Education

Tsinghua University

B.S. in Theoretical and Applied Mechanics

Beijing, China

Sep. 2023 - Present

- **Tsien Excellence in Engineering Program**
- **GPA: 3.90/4.00**
- **Core Academic Training:** Finite Element Method, Physics-based Simulation, Viscous Fluid Mechanics, Solid Mechanics, Convex Optimization

Research Experience

Research Intern

2026/07 – Present

Harvard University

Cambridge, MA, USA

Advisor: Prof. Katia Bertoldi

- Studied rubber infusion-induced buckling and bulging sequences through experiments and simulations.
- Investigated snap-through dynamics of arches and shells using numerical simulations and modal analysis.
- Investigated the structural-acoustic mechanisms of sound generation during snap-through of arches and shells.

Research Intern

2025/09 – 2026/08

Tsinghua University

Beijing, China

Advisor: Prof. Weixi Huang

Project: Flapping-enhanced active olfactory sampling in mosquitoes through near-field odor transport

- Developed a three-dimensional flow-odor coupled solver and introduced a jump-condition correction scheme for complex boundary.
- Quantified how flapping-induced vortex reshapes near-field scalar transport and enhance odor sampling.
- Developed a physics-based transfer learning neural network to investigate the inverse problem of odor-source navigation.
- **First-author manuscript under review at *Bioinspiration & Biomimetics*.**

ISURE Program

2025/07 – 2025/12

University of Notre Dame

Notre Dame, IN, USA

Advisor: Prof. Jianxun Wang

- Investigated neural field models for dimensionality reduction of turbulence data, contributed to the manuscript **Conditional Neural Field for Spatial Dimension Reduction of Turbulence Data: A Comparison Study**, accepted in *Physics of Fluids*.
- Explored reinforcement learning strategies for bio-inspired locomotion and high-order PDE control, like Kolmogorov-Sivashinsky equation.
- Conducted PDE closure modeling to understand nonlinear dynamics and structures.

Research Intern

2024/09 – 2025/09

Tsinghua University

Beijing, China

Advisor: Prof. Hong Chen

Project: Real-Time Event-Based Daily Human Action Recognition with On-Chip Learning on ANP-Is

- Constructed a low-latency event-processing pipeline integrating denoising and neuromorphic encoding for robust action recognition.
- Implemented hardware-software co-design and FPGA-based data transmission, mainly communication protocols
- **First-author demo paper accepted to ISCAS 2026**; presented a live demonstration in Shanghai, May 2026.

Awards

2023	Excellent Presentation Award , 13th Creative Competition & 2023 Tsinghua Student Creative Challenge	Tsinghua University
2024	Comprehensive Excellence Scholarship , Department-level Merit-based Scholarship	Tsinghua University
2024	Tsinghua Technology Taihu Scholarship , For outstanding students in science and technology innovation	Tsinghua University
2025	Tsinghua Technology Taihu Scholarship , For outstanding students in science and technology innovation	Tsinghua University